

Comprehensive: equations + data + word problems

Across Unitcomprehensivereview · T3 equation × T7 applicationquestion × T10 statisticsdata · 65 minutes

Corresponding textbooks: "New Thinking in Primary School Mathematics (Second Edition)" Volume 5 B Units 3 + 7 + 10 + Modern Education 5 B Comprehensive Exercises

Core Traps: |||SEP||SSPA related:

● **Extremely High Frequency** Cross-unit comprehensive questions in paper two, accounting for about 12-18% Prerequisite knowledge:

Classes 14-15 (algebraic expressions + simple equations), Classes 21-31 (data processing + bar graphs), Classes 7-10 (fraction operations) Objectives of this class:

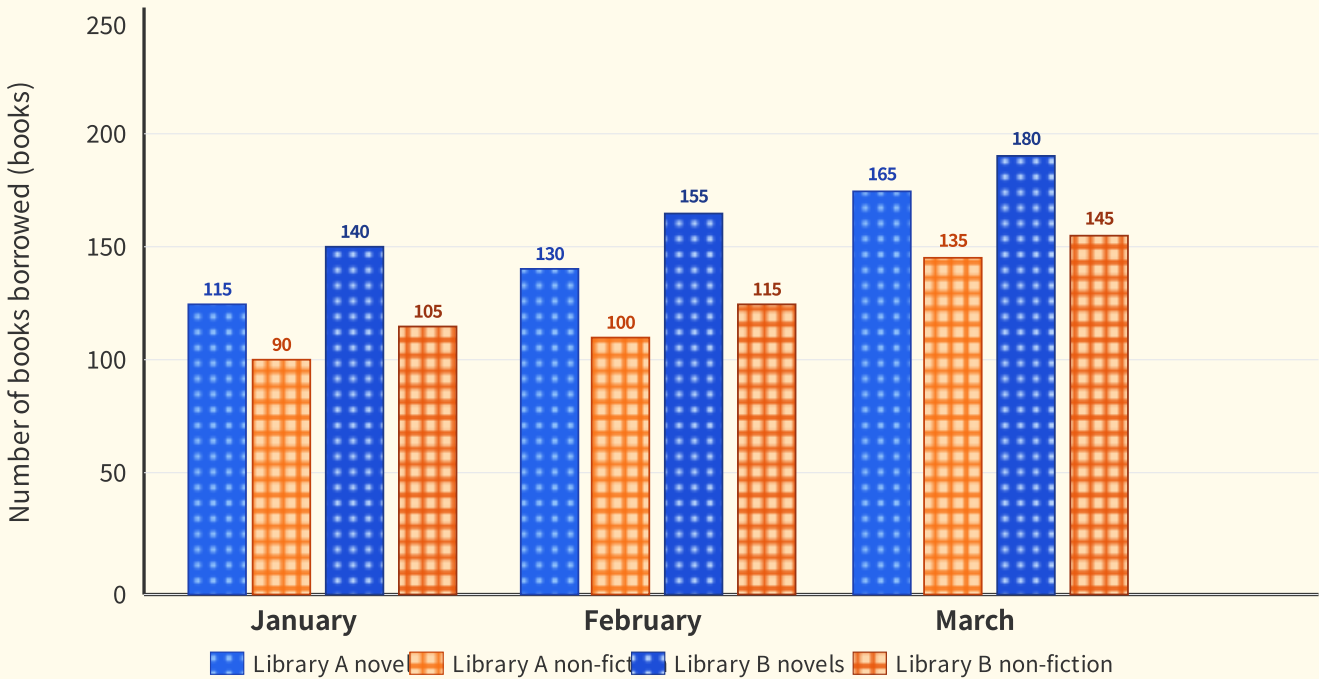
① Establish equations from chart data ② Use equations to solve data analysis problems ③ Comprehensive application across units ① Create equations from graph data ② Use equations to solve data analysis problems ③ Comprehensive application across units

Student Name: _____ Class: _____ Date: _____ Time Spent: _____

I.Warm-Up Questions (total 6 question , 5 minutes)

#	Question	Difficulty	Working Space (Show full working)						
1	Solve the equation: $x + 28 = 75$ $x = ?$	Basic							
2	Solve the equation: $5y = 120$ $y = ?$	Basic							
3	In the bar chart below, the height of bar A = 15 and the height of bar B = 12. How much taller is A than B? How many times is the height of A than B? (assuming each square = 3 units)	Basic							
4	3 times a number plus 12 equals 45. Let this number be n and list the equation.	Advanced							
5	The table below shows the two-day sales. If the total sales for the two days are 750 yuan, how many yuan were sold on the second day? <table><tr><th></th><th>Day 1</th><th>Day 2</th></tr><tr><td>Sales (\$)</td><td>320</td><td>?</td></tr></table>		Day 1	Day 2	Sales (\$)	320	?	Advanced	
	Day 1	Day 2							
Sales (\$)	320	?							
6	According to the reference chart below: Store A' s sales are 45 more than Store B' s. If the sales volume of store B is b , write the algebraic expression of the sales volume of store A.	Advanced							

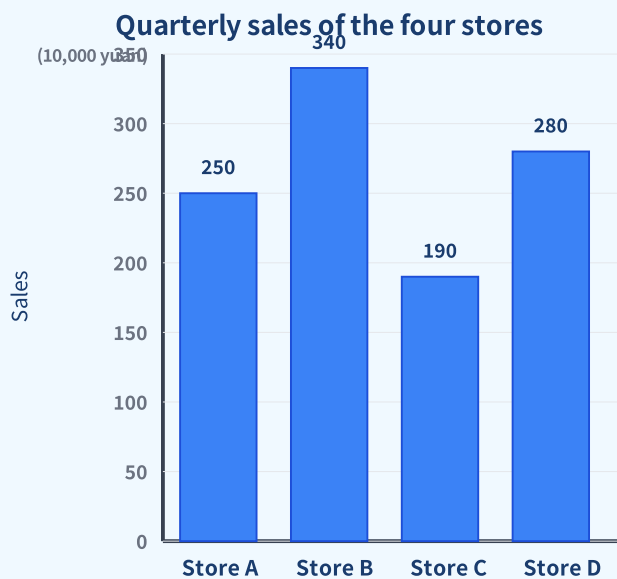
Reference chart: Composite bar chart of library quarterly borrowing statistics



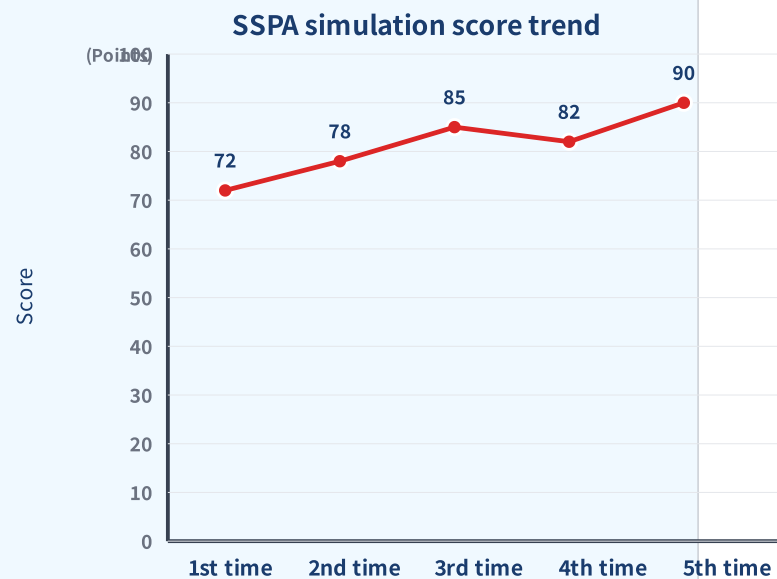
▲ The above is the main reference chart for this class (library borrowing statistics). Most of the sample questions are based on this data.

Data Analysis Reference Chart – Equation Application

Bar chart: comparing different categories



Line chart: Track trends



▲ Exercises in this class: Use equations to solve data analysis problems. For example: "How much more does store A sell than store B?" → Assume the unknowns → Make an equation → Solve the problem.

II. Core Knowledge + Worked Examples

Knowledge point 1: Establishing equations from chart data ● SSPA Core

Four-step method from charts to equations:

- ① **Read charts |||SEP|||:** Extract known values (numbers + units) from bar graphs/data tables. Set unknowns |||SEP|||: Set required (or missing) quantities into variables (such as x , y , n)
- ② **Find equivalence relationships:** "Total", "difference", "several times" and "average" in the question → Convert into equations
- ③ **List of equations:** Write a complete equation using known numbers + variables, ensuring that both sides of the equal sign have equal meanings
- ④ **series of equations:** Write a complete equation using known numbers + variables, ensuring that both sides of the equal sign have equal meanings

□ Trap detonation example (the most important demonstration in this class)

According to the reference chart: It is known that the borrowing volume of library A in January is 115 books for novels and 90 books for non-fiction. If the total number of books borrowed by Library A in January is 20 more than the total number of books borrowed by Library B, what is the total number of books borrowed by Library B in January? Assume parallel equations with unknown variables.

✗ Common mistakes (reversed direction of equation)

Suppose the total quantity of library B
 $= y$
 $y + 20 = 115 + 90$
 $\rightarrow y = 185$ (error: this is the total amount of Hall A)

"A has 20 more than B" was mistakenly written as " $y + 20 = A$ ". It should be " $A = y + 20$ " or " $A - y = 20$ "

✓ Correct solution

The total volume of Hall A in January =
 $115 + 90 = 205$
 Let the total volume of library B in one month = y
 $205 - y = 20 \rightarrow y = 185$
 Or: $205 = y + 20 \rightarrow y = 185$

"A has 20 more than B" $\rightarrow A - B = 20 \rightarrow 205 - y = 20$

 **Tips: "Read the graph to get the numbers, set up the unknowns, find relationships and write equations, solve the equations to check the answers, and bring back the original questions to check the units."**

 **The most frequent error: "A is more than B" is correct to write " $B + n = A$ ", but many students write " $A + n = B$ " instead!**

 **The second most frequent error: looking at the wrong bar (without looking at the legend) when taking numbers from the chart, resulting in the known numbers in the equation being wrong from the beginning!**

Knowledge Point - Worked Examples

#	Question	Difficulty	Working Space
Example 1	According to the reference chart: there were 115 novels in Library A and 140 novels in Library B in January. Suppose the total number of novel books borrowed from the two libraries in January is T, write the expression of T and calculate T.		
Example 2	According to the reference chart: March Jiaguan has 30 more novels than non-fiction. Suppose the number of non-fiction books borrowed by library A in March is n, and use the known number of novel books borrowed (165) to formulate an equation.		

Knowledge Point 1 Synchronization practice

#	Question	Difficulty	Working Space						
7	According to the reference chart: there were 155 novels and 115 non-fiction books in Library B in February. Suppose the total borrowing volume of library B in February is S, write the expression of S and calculate it.								
8	According to the reference chart: the total number of books borrowed from Library A in January was 205. If library B's total monthly volume is 20 less than library A's, let library B's total monthly volume be y, list the equations and solve them.								
9	According to the reference chart: How many times more novels (165 books) in Library A than non-fiction books in Library B (145 books) in March? If the multiple is set to k, list the equation.								
10	The table below shows the sales of the two stores. Total known sales are \$850. If store A's sales are 130 yuan more than store B's, assume store B's sales = b and find b. <table><tr><th colspan="2">Store A</th><th>Store B</th></tr><tr><td>Sales (\$)</td><td>b+130</td><td>b</td></tr></table>	Store A		Store B	Sales (\$)	b+130	b		
Store A		Store B							
Sales (\$)	b+130	b							
11	According to the reference chart: Is the sum of the number of borrowed novels from Library A in January and February more or less than the number of novels borrowed from Library A in March? How much difference? Let the phase difference be d and write the equation.								

Knowledge point 2: Use equations to solve data problems ● SSPA required exam

The complete process of solving data problems with equations:

- ① **Extract data** |||SEP|||: Find all known numbers from charts/tables Set unknown numbers |||SEP|||: Set the quantities you are asked to ask for in the problem as variables
- ② **Build equations**: Based on the relative words in the question (total, difference, multiple, average, remainder...), use known numbers and variables to write equations
- ③ **Solve equations**: Use the principle of the balance (do the same operation on both sides of the equal sign) to find the variables
- ④ **Check calculation + answer**: Substitute into the original question to check whether it is reasonable, and write a complete answer
- ⑤ **Check + Answer**: Substitute into the original question to check whether it is reasonable and write a complete answer.

1

Extract data

Read all known numbers from the chart

2

Let the unknowns

The required quantity is set to x / y / n

3

Build equations

Write an equation using relative words

4

solution + test

Solving the balance principle and substituting it into the verification calculation

example

Example 3: According to the reference chart, the total number of novels borrowed by library A in three months (January to March) = $115 + 130 + 165 = 410$ books. If the number of books borrowed by Library A in March is 3 times a certain number x plus 15, find x .

example

Example 4: The table below shows test scores. It is known that the average score of the four people is 80, find Xiaomei's score m .

	Xiaowen	Xiaomei	Xiaohua	Xiaoqiang
Score	85	m	72	91

⚠ Equation trap: When doing operations on both sides of an equation, you must "do the same operation on both sides at the same time", otherwise the equation will not hold!

⚠ Average trap: average = sum $\div$ number. When formulating equations, we often forget to multiply back the number.
Average score 80 Four people $\rightarrow$ Total = 80×4 !

Knowledge Point 2 Synchronization practice

#	Question	Difficulty	Working Space
12	According to the reference chart: Library B's total borrowing volume of non-fiction books in three months = $105 + 115 + 145 = 365$ books. If there are x more novels (180) in March Yi Library than non-fiction, find x .	🌟	

13	According to the reference chart: What fraction of the non-fiction books (90 books) in Library A in January is the non-fiction books in Library B (105 books) in January? Write the fraction and reduce it.										
14	Four times a number minus 28 equals the number of novels borrowed by Library A in February (130). Let this number be n, list the equation and solve it.										
15	According to the reference chart: the combined number of novels in both libraries in January is 255. If the total number of novels in the two libraries in February is n less than that in January, and the total number of novels in the two libraries in February = 130 + 155 = 285, find n.										
16	The table below shows the total borrowing volume for three months. The average three-month total borrowing volume is known to be 540 books. Let the February total be F, write down the equation and solve it. <table border="1"> <thead> <tr> <th>monthly parts</th><th>January</th><th>February</th><th>March</th></tr> </thead> <tbody> <tr> <td>Total number of books borrowed</td><td>450</td><td>F</td><td>625</td></tr> </tbody> </table>	monthly parts	January	February	March	Total number of books borrowed	450	F	625		
monthly parts	January	February	March								
Total number of books borrowed	450	F	625								

advanceddataequationpractice

#	Question	Difficulty	Working Space
17	Original price of a book \$p. The bookstore offers a 20% discount (i.e. selling price = original price × 0.8). Xiao Ming bought 3 copies and paid a total of \$192. Find the original price p. (Hint: 3 × 0.8p = 192)		
18	According to the reference chart: Let the non-fiction in Library B (115 books) in February be y. If y is 0.7 times less than 5 of a certain Yuejiakan novel, determine which Yuejiakan novel borrowing volume is closest to this value? (115 ÷ 0.7 ≈ 164.3)		

Knowledge Point 3: Comprehensive Application - Equations + Data + Word Questions SSPA Killer Questions

Problem-solving framework for cross-unit comprehensive questions:

- ① **Step 1: Review the question**— Circle all numbers, units, keywords ("Total", "Difference", "Times", "Average", "Remaining")
- ② **Step 2: Extract**— Extract data from charts and conditions from text
- ③ **Step 3: Modeling**— Assume unknowns and convert the conditions into equations/algebraic expressions
- ④ **Step 4: Solve the equation**— Solve the equation and get the value
- ⑤ **Step 5: Extend**— Some questions require recalculation of the solved value (such as finding proportions, percentages, and comparisons)

example

Example 5: According to the reference chart, the total number of novels in Library A in three months is 410. Add this number to the total number of novels borrowed by library B in three months ($140 + 155 + 180$), and then divide it by 6 to get the average monthly borrowing volume of novels in the two libraries (three months). Find this average.

example

Example 6: The article counts the number of people borrowing books from the library in the past four months:

$120 \rightarrow 30$ more than last month $\rightarrow 150 \rightarrow 20$ less than last month

It is known that the first month = 120, the second month = $120 + 30 = 150$, the third month = 150, how many people are there in the fourth month? What is the total number of people in four months?

Knowledge Point 3 Synchronization practice

#	Question	Difficulty	Working Space
19	According to the reference chart: the total non-fiction books in the two libraries in March = $135 + 145 = 280$ books. If this quantity is equal to "5 times a number n plus 30," write the equation and find n .		
20	According to the reference chart: January's total borrowings (all four bars) are more or less than February's total borrowings? Express it in the form of an equation: Suppose the total amount in January = A , the total amount in February = B , and calculate $B - A$.		
21	If among the 180 novels in March Yiguan, $\frac{1}{3}$ is a new book, and the rest are old books. How many new books are there? How many old books are there?		
22	The school counted the number of latecomers for three months: September: a person October: 12 more people than September November: 8 less people than October (a) Write the algebraic expression of the number of latecomers in October and November (b) If the total number of latecomers in three months is 130, find a		

III. Lesson Layered Synchronization Practice

Basic layer (total 4 questions, everyone must do)

#	Question	Difficulty	Working Space
23	According to the reference chart: How many books (fiction + non-fiction) were borrowed in Library A in January?		
24	Solve the equation: $x + 55 = 205$ (Hint: 205 = total number of books borrowed from Library A in January)		
25	According to the reference chart: How many more novels are there in Library B than in Library A in February? Let the phase difference be d.		
26	According to the reference chart: Which Yuejia library has the highest number of borrowed novels? How many copies?		

Advanced layer (total 4 questions, choose do)

#	Question	Difficulty	Working Space												
27	According to the reference chart: What is the total number of books borrowed from the two libraries in March (all four bars added)? Let total = T and calculate T.														
28	The sum of the two numbers is 72 and the difference between the two numbers is 18. Let the smaller number be x and the larger number x + 18. List the equations to find the two numbers.														
29	According to the reference chart: If you add up the March novels in Library A (165) and the novels in March in Library B (180) and divide them by 3, what is the result? Calculate using comprehensive formula.														
30	Based on the reference chart data, complete the table below and calculate the total non-fiction borrowing volume for each library in three months: <table><tr><th></th><th>January</th><th>February</th><th>March</th></tr><tr><td>Library A non-fiction</td><td>90</td><td>100</td><td>135</td></tr><tr><td>Library B non-fiction</td><td>105</td><td>115</td><td>145</td></tr></table> Which library has a higher total non-fiction volume? Write an equation expressing the phase difference.		January	February	March	Library A non-fiction	90	100	135	Library B non-fiction	105	115	145		
	January	February	March												
Library A non-fiction	90	100	135												
Library B non-fiction	105	115	145												

 challenge layer (total 3 questions, choose do)

#	Question	Difficulty	Working Space						
31	There is one missing data in the table below. It is known that the sum of A and B is 240, and B is 40 more than A. Find A and B. <table><tr><th>Item</th><th>A</th><th>B</th></tr><tr><td>Value</td><td>?</td><td>?</td></tr></table> (Hint: Let A = a, then B = a + 40 → 2a + 40 = 240)	Item	A	B	Value	?	?		
Item	A	B							
Value	?	?							
32	Continuing from the previous question: If you also know that "C is 2 times of A but 10 less", and the sum of A, B, and C is 430, find C.								

33	Based on all the data in the reference chart (12 values), answer: (a) What percentage of the total number of borrowed books does novels account for? (Reduction) (b) If the March data is extracted, the total volume of novels in March: the total volume of non-fiction in March = ? : ? (expressed in the simplest integer ratio)		
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IV.comprehensiveapplicationquestion (equation + data + application, all must do)

applicationquestionspecial topic (formula → equation → find solution → check → answer sentence)

#	Question	Difficulty	Working Space								
34	Xiaomei has saved x yuan, and Xiaohua has saved 45 yuan more than her. The two of them saved a total of \$285. How much money has Xiaomei saved?										
35	The length of a rectangle is 8 cm more than its width. The perimeter is 64 cm. Let the width be w cm. (a) Let w represent the length (b) List the perimeter equation (c) Find the length and width in cm?										
36	The following table shows the scores for the three tests: <table border="1"> <thead> <tr> <th>Test</th><th>First time</th><th>Second time</th><th>Third time</th></tr> </thead> <tbody> <tr> <td>Score</td><td>78</td><td>m</td><td>$m - 6$</td></tr> </tbody> </table> It is known that the average score of the three times is 80. Find m (the second fraction) and the third fraction.	Test	First time	Second time	Third time	Score	78	m	$m - 6$		
Test	First time	Second time	Third time								
Score	78	m	$m - 6$								
37	Stationery stores sell pencils and ball pens. Pencils are \$3 each and ball pens are \$5 each. Xiao Ming bought a total of 20 pencils and ball pens, and paid a total of \$76. Suppose p pencils are purchased. (a) How many ball pens did you buy? (expressed in terms of p) (b) List the equation for the total amount (c) Find p .										
38	Admission to the Children's Palace: \$30 for adults, \$15 for students. A total of 200 tickets were sold on one day, with total revenue of \$4500. Suppose s student tickets are sold. (a) How many adult tickets were sold? (Expressed by s) (b) List the total revenue equation (c) Find how many adult and student tickets are sold?										

V. Class afterhomework

Basic must-do questions (total 4 questions)

#	Question	Difficulty	Working Space
H1	According to the reference chart: Library A's non-fiction borrowing volume in January = 90. Let the non-fiction of library B = n in that month. If n is 15 more than 90, write the equation to find n and check the values on the graph.		
H2	According to the reference chart: Calculate the sum of the novels in House A and House B in February.		
H3	Solve the equation: $240 - x = 95$ (Hint: x = difference between January non-fiction in Hall A and January non-fiction in Hall B)		
H4	The father is 3 times as old as the son. Their combined ages are 52. Assume the age of the son is n , what is the age of each of them?		

For advanced, choose doquestion (total 4 questions, choose do)

#	Question	Difficulty	Working Space												
H5	According to the reference chart: the total volume of novels in Library A ($115+130+165=410$) and the total volume of novels in Library B ($140+155+180=475$) in three months. Assume that the number of novels in Library B is d more than that in Library A. Write an equation to find d . If Library A wants to close the gap next season, on average, how many more novels will it borrow per month than Library B? (set to x)														
H6	A sequence of numbers: first number = 35, each subsequent number is 8 greater than the previous one. Let the value of the n th number be $35 + 8(n - 1)$. Find the value of the 5th number. If the k th number = 99, find k .														
H7	Supermarket drink sales statistics (boxes): <table border="1"> <thead> <tr> <th></th><th>Soda</th><th>Juice</th><th>Tea</th></tr> </thead> <tbody> <tr> <td>Last week</td><td>45</td><td>38</td><td>52</td></tr> <tr> <td>This week</td><td>x</td><td>42</td><td>$x+10$</td></tr> </tbody> </table> It is known that the total sales this week are 30 boxes more than last week. Find x and the sales volume of soda and tea this week.		Soda	Juice	Tea	Last week	45	38	52	This week	x	42	$x+10$		
	Soda	Juice	Tea												
Last week	45	38	52												
This week	x	42	$x+10$												
H8	Prices of three books: Book A is \$15 more expensive than book B. Book B is \$10 more expensive than book C. Three books total \$215. Let the price of book C be c. (a) Use c to express the prices of book B and book A (b) List the total price equation (c) Find the prices of each of the three books														

VI. Ultimate challenge area

#	Question	Difficulty	Working Space												
1	School sports meeting statistics: <table><tr><th></th><th>Running</th><th>Long jump</th><th>Softball throw</th></tr><tr><td>Boys (person)</td><td>a</td><td>b</td><td>a-5</td></tr><tr><td>Girls (person)</td><td>b-10</td><td>a+5</td><td>b+8</td></tr></table> Known: Total number of runners = 70, total number of long jump = 75. (a) List two equations (b) Find a and b (c) Find the total number of people throwing the softball		Running	Long jump	Softball throw	Boys (person)	a	b	a-5	Girls (person)	b-10	a+5	b+8		
	Running	Long jump	Softball throw												
Boys (person)	a	b	a-5												
Girls (person)	b-10	a+5	b+8												
2	Based on all the data in the reference chart, reorganize a total data table and calculate: (a) The total number of borrowed books for all novels in three months (b) The total number of borrowed books for all non-fiction in three months (c) How many times is the total number of novels than the total number of non-fiction? (Answers expressed as decimals, rounded to one decimal place) (d) If the total number of novels and non-fiction is to be equal, how many more non-fiction books would need to be borrowed? (set to z)														

VII. The Lessoncorecommon errorssummary

Self-examination in this hall (tick after completion)

- ☐ I know the pitfalls of solving each knowledge point
- ☐ I can complete 🦉 basic questions independently
- ☐ I can challenge 🌿 advanced questions
- ☐ I remember the formula

🎯 Review of Learning Objectives - After completing this lesson you should be able to:

- ☐ Identify all trap types in our hall
- ☐ Solve 🦉 basic questions independently (100% correct)
- ☐ Challenge 🌿 Advanced questions (80%+ correct)
- ☐ Explain the lesson formula to classmates

#	common error	Correct Approach
1	The equation "A is more than B" is written in reverse SEP : write "A = B + difference" as "B = A + difference": Write "A = B + difference" as "B = A + difference"	Test with numbers: 10 is 3 more than 7 → 10 = 7 + 3 → large number = decimal + difference
2	The missing multiplication of the average equation SEP : average = sum ÷ number → sum = average × number: Average = sum ÷ number → sum = average × number	When formulating equations: when calculating the sum by knowing the average, you must first multiply by the number
3	Get the wrong number from the chart SEP : don' t look at the legend, look at the wrong month, look at the wrong bar: Not looking at the legend, looking at the wrong month, looking at the wrong bar	Three-step confirmation: legend → month → pattern → read scale
4	The operations on both sides of the equation are not equal SEP : One side is added, the other side is not added: One side is added, the other side is not added	Principle of the balance: both sides perform exactly the same operation at the same time
5	Algebraic bracket handling error SEP : For example, 3(x + 8) only multiplies 3x and forgets to multiply 3×8: For example, 3(x + 8) only multiplies 3x and forgets to multiply 3×8	Distributive law of multiplication: a(b + c) = ab + ac, each item must be multiplied

6	The units are inconsistent when solving the problem SEP : the chart uses "this", the question uses "book", or the conversion of Yuan/Jiao is wrong: "Ben" is used in the chart, "book" is used in the question, or the conversion of Yuan/Jiao is wrong.	Formulate equations after unifying the units; answer in correct units
7	Missing verification step SEP : No substitution check after solving x: No substitution check is performed after solving x	Substituting x into both sides of the original equation, confirm that the left side = the right side before writing the answer